

Automated Research Report Generator using n8n

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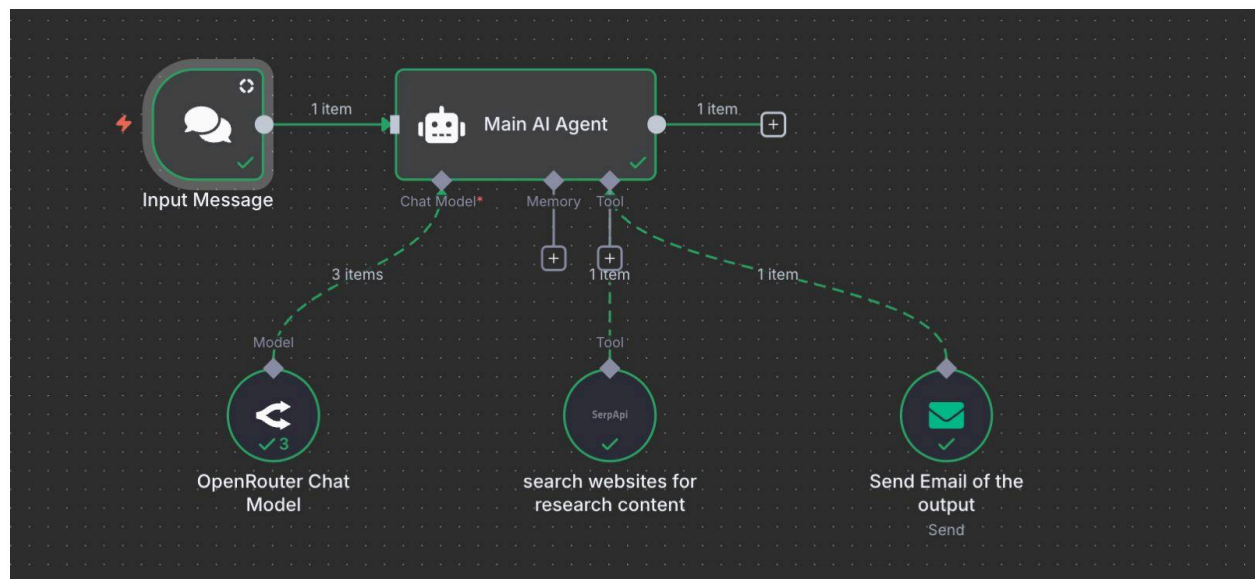
Student ID: 23155082

Course: CSO-314: Powers of AI

Project Title: Automated Research Report Generator (Option2)

Submission Date: 10th March, 2026

N8N Workflow Screenshot -



N8N Workflow JSON File:

```
{ "name": "Automated Research Report Generator",

  "nodes": [

    {

      "parameters": {

        "public": true,

        "initialMessages": "Hi there! 🙋\nWhat do you want to Research???",

        "options": {}

      },

      "type": "@n8n/n8n-nodes-langchain.chatTrigger",

      "typeVersion": 1.3,

      "position": [

        416,

        0

      ],

      "id": "7b53e8dd-c3a8-4577-b02c-804931ae9ec0",

      "name": "Input Message",

      "webhookId": "fa989028-3cb5-47d6-a197-6509ae2aac0c",

      "alwaysOutputData": true,

      "notes": "Input data for the AI Agent"

    },

    {

      "parameters": {
```

```
    "model": "nvidia/llama-3.3-nemotron-super-49b-v1.5",

    "options": {

        "temperature": 0.7

    }

},

"type": "@n8n/n8n-nodes-langchain.lmChatOpenRouter",

"typeVersion": 1,

"position": [

    480,

    256

],

"id": "c8c883ce-20b1-477a-b656-5305e7b11f16",

"name": "OpenRouter Chat Model",

"credentials": {

    "openRouterApi": {

        "id": "YyXskfaF0bjcH0tZ",

        "name": "OpenRouter account"

    }

},

"notes": "Research"

},

{

    "parameters": {

        "descriptionType": "manual",
```

```
"toolDescription": "send the output in text format as the email body",

"fromEmail": "priyanka.rhichkad.min23@itbhu.ac.in",

"toEmail": "priyanka.rhichkad.min23@itbhu.ac.in",

"subject": "=({ / *n8n-auto-generated-fromAI-override*/ $fromAI('Subject', ``,
'string') }}",

"emailFormat": "both",

"text": "=({ / *n8n-auto-generated-fromAI-override*/ $fromAI('Text', ``,
'string') }}",

"html": "=({ / *n8n-auto-generated-fromAI-override*/ $fromAI('HTML', ``,
'string') }}",

"options": {}

},

"type": "n8n-nodes-base.emailSendTool",

"typeVersion": 2.1,

"position": [

  1056,

  256

],

"id": "36c5a490-e284-43d8-999d-0da8ec28555d",

"name": "Send Email of the output",

"webhookId": "ca49fcb6-db64-4464-9755-0a29dfce7882",

"credentials": {

  "smtp": {

    "id": "zTNvt2Yhfvr2j9F",

    "name": "SMTP account 4"
```

```
    }

    },

    "notes": "Use this tool to send the final research report via email to the user after the report is generated.\n\nyou could write in the format-\nHello,\n\nHere is the AI generated research report you requested on the chatInput.\n\n{The output from the model}\n\nBest regards,\nPriyanka's AI Agent"

  },

  {

    "parameters": {

      "options": {}

    },

    "type": "@n8n/n8n-nodes-langchain.toolSerpApi",

    "typeVersion": 1,

    "position": [

      768,

      256

    ],

    "id": "90ae1572-bac6-473b-b47c-e689c63bcf32",

    "name": "search websites for research content",

    "credentials": {

      "serpApi": {

        "id": "UygLLzXPDqYOcXaP",

        "name": "SerpAPI account 3"

      }

    },

  },

}
```

```
    "notes": "Use this tool to search the internet for articles, research papers, and webpages about a given topic. \nReturn 3-5 relevant results with titles and links."

  },

  {

    "parameters": {

      "promptType": "define",

      "text": "=Generate a research report on the following topic:\n\n{{$json.chatInput }}\n\nInstructions:\n1. Use the Research Agent tool to gather information from the internet.\n2. Create a structured research report based on the findings from at least 3 to 5 relevant articles or web pages in 300-400 words.\n3. After generating the report, use the Send Email tool to email the report to the user in plain text format without Markdown formatting.",

      "options": {

        "systemMessage": "=You are the main AI workflow agent responsible for generating and delivering research reports.\n\nYou have access to two tools:\n\n1. Web Search Tool\nThis tool performs internet research using SerpAPI and returns summarized findings with sources.\n\n2. Send Email Tool\nThis tool sends an email containing the final research report.\n\nYour task is to coordinate the workflow as follows:\n\nStep 1: When a user provides a topic, call the Web search tool to gather reliable information about the topic.\n\nStep 2: Review the research results returned by the tool and convert them into a clear, structured research report.\n\nStep 3: The report must include:\n- Title\n- Introduction\n- Key Insights\n- Current Applications\n- Future Outlook\n- Conclusion\n- Sources and links\n\nThe report should be approximately 300-400 words and from at least 3 to 5 relevant articles or web pages.\n\nStep 4: After generating the report, use the Send Email tool to send the final report to the user in plain text format without Markdown formatting.\n\nImportant Rules:\n- Always use the Research Agent tool before writing the report.\n- Always use the Send Email tool after generating the report.\n- Do not invent facts. Only use information gathered from the research tool.\n- Keep the language clear, professional, and informative."

      }

    },

    "type": "@n8n/n8n-nodes-langchain.agent",

    "typeVersion": 2.2,
```

```
    "position": [

        624,

        0

    ],

    "id": "6adb49e3-99e9-43d5-b6ea-909c87e05baf",

    "name": "Main AI Agent"

}

],

"pinData": {},

"connections": {

    "Input Message": {

        "main": [

            [

                {

                    "node": "Main AI Agent",

                    "type": "main",

                    "index": 0

                }

            ]

        ]

    },

    "OpenRouter Chat Model": {

        "ai_languageModel": [

            [
```

```
{
  "node": "Main AI Agent",
  "type": "ai_languageModel",
  "index": 0
}

]

],

},

"Send Email of the output": {
  "ai_tool": [
    [
      {
        "node": "Main AI Agent",
        "type": "ai_tool",
        "index": 0
      }
    ]
  ]
},

"search websites for research content": {
  "ai_tool": [
    [
      {
        "node": "Main AI Agent",
```

```
        "type": "ai_tool",

        "index": 0

    }

]

]

},

"Main AI Agent": {

    "main": [

        []

    ]

}

},

"active": false,

"settings": {

    "executionOrder": "v1"

},

"versionId": "c6441256-dd8f-44b7-af9a-2b4918f30f91",

"meta": {

    "templateCredsSetupCompleted": true,

    "instanceId": "c76cbd41aac0dfc1d3fc04431354ea8db1c7d28d2aab6034aa40ae73cdec24d6"

},

"id": "spkARo34aOHq9xTI",

"tags": []

}
```

Video Demonstration (Google Drive link):

https://drive.google.com/file/d/14u49_9pDViZAKX2AlbsEjt0A81hOOLlr/view?usp=drive_link

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Introduction:

In today's digital world, a huge amount of information is available on the internet. Whenever I want to learn about a new topic or conduct research, I usually need to go through multiple websites, articles, and resources to collect relevant information. This process can be time-consuming and sometimes overwhelming because the information is scattered across many sources.

To address this problem, I decided to build an **AI-powered research assistant** that can automatically search for information on a given topic and generate a structured research report. The idea behind this project is to reduce the manual effort required to search, read, and summarize information from different websites.

For this project, I used n8n to design an automated workflow that integrates AI agents with external tools. The system takes a research topic as input, searches the internet for relevant resources, extracts key insights, and generates a summarized report. Finally, the report is automatically sent to the user's email.

I chose this problem because research and information gathering are common tasks for students, professionals, and researchers. Automating this process using AI agents demonstrates how workflow automation and artificial intelligence can work together to save time and improve productivity. This project also helped me understand how AI agents interact with tools, perform reasoning, and complete multi-step tasks in an automated workflow.

Workflow Design:

The workflow is designed to simulate the behaviour of an intelligent research assistant. It consists of several interconnected nodes that perform different tasks in the automation pipeline.

1. Input Message Node

The workflow begins with an Input Message node, which serves as the system's entry point. In this step, the user provides the research topic. This topic will serve as the primary input for the AI agent's analysis.

For example, a user may input a topic such as *“Future of quantum computing”*. This input is then passed to the AI agent for further processing.

2. AI Agent

The AI Agent acts as the central controller of the workflow. Its primary responsibility is to interpret the user's request and decide how the workflow should proceed. Instead of performing the research directly, the main agent calls a specialised research agent tool to gather relevant information.

The main agent uses a chat model provided through **OpenRouter**, which enables access to different large language models. The AI model processes prompts, understands the research topic, and coordinates the workflow.

The prompt provided to the agent instructs it to generate a structured research report that includes the following sections:

- Title
- Introduction
- Key Insights
- Current Applications
- Future Outlook
- Conclusion
- Sources

The report is generated in plain text format so that it can be easily delivered through email.

3. AI Chat Model

The reasoning and text generation capabilities of the AI agents are powered by a chat model connected through **OpenRouter**. The chat model enables the agents to understand the user's input, analyze information retrieved from external sources, and generate structured responses.

In this workflow, the chat model performs several important tasks:

- Interpreting the research topic provided by the user
- Deciding when to use available tools such as the web search API
- Summarizing information collected from different sources
- Generating a structured research report

The use of a chat model allows the workflow to simulate the behavior of an intelligent research assistant that can analyze information and produce meaningful insights.

4. Web Search Tool

The research agent uses **SerpAPI** to perform automated web searches. This tool allows the agent to retrieve information from online sources such as articles, blogs, and documentation pages.

The research agent performs the following actions:

- Searches for relevant content related to the topic
- Identifies useful articles and resources
- Extracts important insights from these sources

Using a search API ensures the generated research report includes up-to-date information rather than relying solely on the model's training data.

5. Email Tool

After the research report is generated, the output is sent to the **Send Email node**. This node automatically delivers the generated report to the user's email address.

This final step completes the automation pipeline by ensuring that the research output is directly shared with the user without requiring manual copying or formatting.

Overall, the workflow integrates AI reasoning, web search capabilities, and communication tools to automate the process of research report generation.

Challenges & Solutions:

During the development of this workflow, several technical challenges were encountered.

AI Agent Not Using Tools - Initially, the AI agent generated responses directly from the chat model rather than calling the search tool. This prevented the workflow from retrieving real-time information from the internet.

This issue was resolved by improving the **system prompt and tool descriptions**. The instructions were modified to clearly state that the agent must use the search tool to gather research information before generating the final report.

API Authentication Issues - Another challenge involved configuring the API credentials required for external services. Incorrect API configurations caused authentication errors when connecting to certain services.

To resolve this issue, the API credentials were carefully configured and verified within the workflow platform. Once the correct credentials were applied, the tools successfully connected to the required services.

Email Formatting Issues - During testing, the generated research report was sent in **Markdown format**, which caused formatting symbols such as asterisks and headings to appear in the email.

This issue was resolved by modifying the system prompt and instructing the AI agent to return the final report in plain text rather than Markdown. This ensured that the email content appeared clean and readable.

Conclusion:

This project successfully demonstrates how AI agents can be combined with workflow automation to create a powerful automated research assistant. Using n8n, the system integrated AI reasoning, web search capabilities, and automated email delivery into a single workflow.

The system accepts a research topic as input, searches the internet using the SerpAPI, analyses the results using AI models via OpenRouter, and generates a structured research report. The final output is then automatically delivered to the user through email.

This automation significantly reduces the time required to gather and summarise research information. It also demonstrates the potential of AI agents to perform complex multi-step tasks by interacting with external tools.

In the future, this workflow could be improved by integrating additional tools such as webpage content extractors, academic research APIs, or document storage systems. These improvements would allow the system to generate more detailed reports and store research outputs for future reference.

Overall, the project highlights how AI-driven workflow automation can simplify information gathering and improve productivity in research-related tasks.